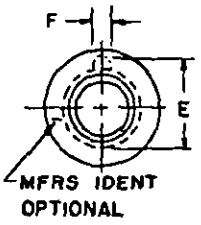
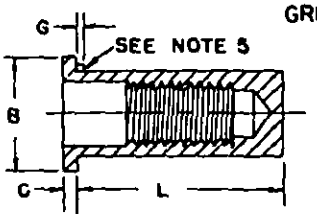
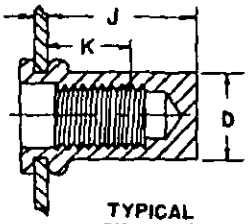
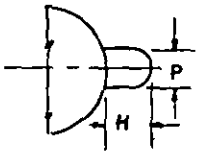
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TYPICAL INSTALLATION



KEYWAY DETAIL

TABLE I

DASH NUMBER STL	DASH NUMBER AL ALY	DASH NUMBER CRES	THREAD SIZE	GRIP RANGE MIN	GRIP RANGE MAX	B HEAD DIA ±.015	C HEAD HEIGHT NOM	D SHANK DIA +.000 -.004	E KEY HEIGHT MAX	F KEY WIDTH +.005 -.000	G KEY LENGTH MAX	KEYED L SHANK LENGTH ±.015	KEYED J INSTL LENGTH REF	KEYED K STUD MAX DEPTH REF	KEYLESS L SHANK LENGTH ±.015	KEYLESS J INSTL LENGTH REF	KEYLESS K STUD MAX DEPTH REF
1	2			.010	.060							.500	.385	.230	.500	.385	.230
3	4		4-40	.060	.085	.270	.025	.155	.198	.054	.023	.525	.385	.230	.525	.385	.230
5	6		UNC-3B	.085	.110							.555	.390	.230	.555	.390	.230
7	8			.010	.075							.750	.615	.405	.625	.490	.305
9	10		6-32	.075	.120	.325	.032	.189	.240	.054	.023	.750	.565	.355	.625	.440	.255
11	12		UNC-3B	.120	.160							.750	.520	.310	.750	.520	.260
13	14			.010	.075							.750	.615	.405	.625	.490	.305
15	16		8-32	.075	.120	.357	.032	.221	.271	.054	.023	.750	.565	.355	.625	.440	.255
17	18		UNC-3B	.120	.160							.750	.520	.310	.750	.520	.260
19	20	C85	10-32	.010	.080							.781	.630	.380	.781	.630	.380
21	22			.080	.130	.406	.038	.250	.302	.054	.023	.843	.630	.390	.843	.640	.390
23	24		UNF-3B	.130	.180							.891	.640	.390	.891	.640	.390
25	26			.020	.080							.937	.760	.440	.937	.760	.440
27	28		1/4-28	.080	.140	.475	.058	.332	.382	.054	.035	1.000	.760	.440	1.000	.760	.440
29	30		UNF-3B	.140	.200							1.062	.760	.440	1.062	.760	.440
49	50	C55		.020	.080							.937	.760	.440	.937	.760	.440
51	52	C57	1/4-20	.080	.140	.475	.058	.332	.382	.054	.035	1.000	.760	.440	1.000	.760	.440
53	54	C59	UNC-3B	.140	.200							1.062	.760	.440	1.062	.760	.440
31	32			.030	.125							1.287	.940	.550	1.187	.940	.550
33	34		5/16-24	.125	.200	.665	.062	.413	.505	.120	.040	1.281	.960	.555	1.281	.960	.555
35	36		UNF-3B	.200	.275							1.343	.950	.560	1.343	.950	.560
61	62	C67		.030	.125							1.187	.940	.550	1.187	.940	.550
63	64	C69	5/16-18	.125	.200	.665	.062	.413	.505	.120	.040	1.281	.960	.555	1.281	.960	.555
65	66	C71	UNC-3B	.200	.275							1.343	.950	.560	1.343	.950	.560
37	38			.030	.115							1.281	1.020	.660	1.281	1.020	.660
39	40		3/8-24	.115	.200	.781	.088	.490	.597	.120	.040	1.375	1.030	.670	1.375	1.030	.670
41	42		UNF-3B	.200	.285							1.468	1.040	.680	1.468	1.040	.680
73		C79		.030	.115							1.281	1.020	.660	1.281	1.020	.660
75		C81	3/8-16	.115	.200	.781	.088	.490	.597	.120	.040	1.375	1.030	.670	1.375	1.030	.670
77		C83	UNC-3B	.200	.285							1.468	1.040	.680	1.468	1.040	.680
43	44			.025	.145							1.656	1.325	.855	1.656	1.325	.855
45	46		1/2-20	.145	.265	1.000	.125	.640	.772	.151	.040	1.781	1.330	.865	1.781	1.330	.865
47	48		UNF-3B	.265	.385							1.906	1.335	.875	1.906	1.335	.875

Ⓒ INACTIVE FOR NEW DESIGN AFTER 18 APRIL 1973

Ⓒ For changes see sheets 1 and 2.

REVIEWER:
USER:

This military standard is mandatory for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

APPROVED 21 MAR 61 REVISED (A) 9 AUG 61 (B) 3 JUL 1968 (C) 18 APR 1973

P.A. WC	TITLE	MILITARY STANDARD
Other Cont AS 82	NUT, PLAIN, BLIND RIVET-FLAT HEAD, CLOSED END	MS 27131
PROCUREMENT SPECIFICATION NONE	SUPERSEDES:	SHEET 1 OF 2

DD FORM 672-1 (Continued) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

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5310

TABLE II

SIZE	HOLE SIZE		DRILL SIZE	* AVERAGE ULTIMATE STRENGTH						KEYWAY DIMENSIONS		
				TENSION LB			SINGLE SHEAR LB			P	H	
	MIN	MAX		AL ALY	STEEL	CRES	AL ALY	STEEL	CRES	±.003	MIN	MAX
NO. 4	.155	.157	5/32	261	458	--	138	316	--	.062	.046	.048
NO. 6	.189	.193	NO. 12	373	656	--	230	460	--	.062	.056	.058
NO. 8	.221	.226	NO. 2	485	853	--	294	588	--	.062	.056	.058
NO. 10	.250	.256	E	617	1085	--	374	748	--	.062	.056	.058
1/4	.332	.338	Q	1150	1850	2510	710	1100	1790	.062	.056	.058
5/16	.413	.423	Z	1600	2750	3730	930	1750	2850	.128	.097	.102
3/8	.490	.500	12.5MM	2075	3900	5280	1260	2420	3940	.128	.110	.115
1/2	.640	.650	41/64	3500	6620	--	2000	4080	--	.159	.135	.140

* Average strength of random samples shall be equal to the values shown in this table within minus fifteen percent.

NOTES:

1. MATERIAL: Steel, C1108 or C1110 per Fed-Std-66. (C)
   Aluminum alloy per QQ-A-430, type 6053-T4 (except temper).
   Corrosion resisting steel per Fed-Std-66, composition 430.
2. PROTECTIVE FINISH: Steel, cadmium plated, per QQ-P-416, type II, class 3.
   Aluminum alloy, anodized per MIL-A-8625, type II.
   Corrosion resisting steel, passivated per QQ-P-35.
3. THREADS: Threads shall be in accordance with MIL-S-7742.
4. DIMENSIONS: Dimensions are in inches.
5. PART NUMBER: The MS part number consists of the MS number, plus the dash number.
   Example: MS27131-1. When key type is specified, add K to dash number.
   Example: MS27131-1K.
6. Reference documents shall be of the issue in effect on date of invitations for bid.
7. Special pull-up tools are required for proper installation.
8. Dimensions shown for "K" represent the maximum penetration of the attaching stud or capscrew that is recommended. Exceeding these dimensions may cause bottoming.
9. Sheet 2 was added as part of revision B.

(C) INACTIVE FOR NEW DESIGN
AFTER 18 APRIL 1973

REVIEWER:
USAF:

APPROVED 21 MAR 1961 REVISED (A) 9 AUG 1961 (B) 3 JUL 1968 (C) FOR CHANGES SEE SHEETS 1 AND 2.

This military standard is mandatory for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

P.A. WC Other Cust AS A2	TITLE NUT, PLAIN, BLIND RIVET - FLAT HEAD, CLOSED END	MILITARY STANDARD MS 27131
PROCUREMENT SPECIFICATION NONE	SUPERSEDES:	SHEET 2 OF

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT, ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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